



StudentHub

Requirements Analysis and System Design

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1 Introduction

1.1 Purpose

The purpose of StudentHub is to provide a platform by students for students. Its objective is to streamline the management of student life. The platform will allow students to manage their tasks and organise their study sessions. The platform will also provide a space for students to connect with each other, share resources and collaborate on projects. The platform will also offer students a network forum where they can make announcements to find used books and private lessons, as well as share information about school events.

1.1.1 Goals

ID	Description
G1	Allow users to create, delete, log in to and log out from an account
G2	Enable users to create, edit, move, and delete deadlines on a shared calendar
G3	Provide functionality for users to create, edit, and delete private deadlines visible only to themselves
G4	Allow users to upload, share, edit, and delete study notes on various subjects
G5	Enable users to create, manage, and delete study groups with other platform users
G6	Provide a messaging system for communication within study groups
G7	Allow users to send and receive study group invitations
G8	Provide a forum where users can create, edit, and delete announcements
G9	Enable users to comment on and interact with forum announcements
G10	Allow users to search for specific topics, notes, or announcements within the platform
G11	Provide a marketplace functionality for exchanging used academic books
G12	Enable users to offer and search for private tutoring lessons
G13	Allow users to create and share information about upcoming school events
G14	Provide notification functionality for deadline reminders and forum activity
G15	Enable users to customize their profile and privacy settings

Table 1: Goals

1.2 Scope

1.2.1 World Phenomena

ID	Description
WP1	A student wants to create an account
WP2	A student wants to delete an account
WP3	A student wants to create a deadline on the shared/private calendar
WP4	A student wants to move a deadline on the shared/private calendar
WP5	A student wants to edit a deadline on the shared/private calendar
WP6	A student wants to delete a deadline on the shared/private calendar
WP7	A student wants to upload his/her study notes on a given subject
WP8	A student wants to share his/her study notes on a given subject
WP9	A student wants to edit his/her study notes on a given subject
WP10	A student wants to delete his/her shared notes
WP11	A student wants to create a studying group
WP12	A student wants to invite people to his/her studying group
WP13	A student wants to chat with his/her studying group
WP14	A student wants to delete his/her studying group
WP15	A student wants to publish an announcement on the forum
WP16	A student wants to edit an announcement on the forum
WP17	A student wants to delete an announcement on the forum
WP18	A student wants to comment an announcement on the forum
WP19	A student wants to delete a comment on the forum
WP20	A student wants to share an announcement found on the forum
WP21	A student wants to find documents relative to a topic
WP22	A student wants to buy an academic book
WP23	A student wants to offer private tutoring lessons
WP24	A student wants to purchase a private tutoring lessons
WP25	A student wants to customize his profile

Table 2: World Phenomena

1.2.2 Shared Phenomena

ID	Description
SP1	The student creates an account on the system
SP2	The student deletes the account on the system
SP3	The student uploads a deadline in the system
SP4	The student moves a deadline in the system
SP5	The student edits a deadline in the system
SP6	The student deletes a deadline in the system
SP7	The student uploads the study notes in the system
SP8	The student uses the share function of notes provided by the system
SP9	The student edits the study notes in the system
SP10	The student deletes the study notes from the system
SP11	The student creates a study group in the system
SP12	The student invites people to the study group using the system
SP13	The student chats with the study group through the system
SP14	The student deletes the study group from the system
SP15	The student publishes an announcement on the system forum
SP16	The student edits an announcement on the system forum
SP17	The student deletes an announcement from the system forum
SP18	The student comments on an announcement in the system forum
SP19	The student deletes a comment from the system forum
SP20	The student shares an announcement using the system
SP21	The student searches a topic in the system forum
SP22	The student purchases an academic book through the system
SP23	The student offers private tutoring lessons in the system
SP24	The student books a private tutoring lesson using the system
SP25	The student customizes the profile in the system

Table 3: Shared Phenomena

1.3 Revision History

- Version 1.0 - May 4, 2025 - Initial release of the document.

1.4 Document Structure

This document is structured into 6 sections, each containing subsections that provide detailed information about the system. The sections are as follows:

- **Introduction:** This section provides an overview of the system, its purpose, and goals.
- **Overall Description:** This section describes the system's context, stakeholders, and shared phenomena.
- **Specific Requirements:** This section outlines the specific requirements for the system.
- **Effort Spent:** This section provides a summary of the hours worked on the project.
- **References:** This section lists any references used in the document.

2 Overall Description

2.1 Product Perspective

2.1.1 Scenarios

Scenario 1: An user creates an account and manages his own deadlines

Elir, a first-year Computer Science student, creates an account on StudentHub using his university email. After verification, he adds three academic deadlines to the Calendar: a Programming assignment, a Mathematics exam, and a group project meeting. He marks the group project as "shareable" for future teammates. Elir also adds a private reminder to call home on the weekend. Later, when his Programming assignment deadline is extended, Elir easily updates it in the system.

Scenario 2: An user creates an announcement on the forum

Davide T., an Economics student, searches StudentHub's Forum for a specific Econometrics textbook. Not finding it, he creates an announcement requesting "Advanced Econometrics by Rosiglioni D. - 7th Edition" with his budget. Two days later, Salvo comments on his post offering to sell his copy. They arrange the exchange through direct messaging and meet at the university library. Trevisan then marks his announcement as "Resolved" with a thank-you note.

Scenario 3: An user creates a study group and shares notes

Pantarotto A., a Biology student, uploads his Cell Structure notes to StudentHub's "Study Resources" section with relevant tags. He then creates a "Biology Midterm Prep" study group, making it discoverable to other Biology students. Alessio invites five classmates, three of whom accept immediately. Through the group chat, he shares his notes and the team plans their study sessions for the upcoming midterm.

2.1.2 Domain Model

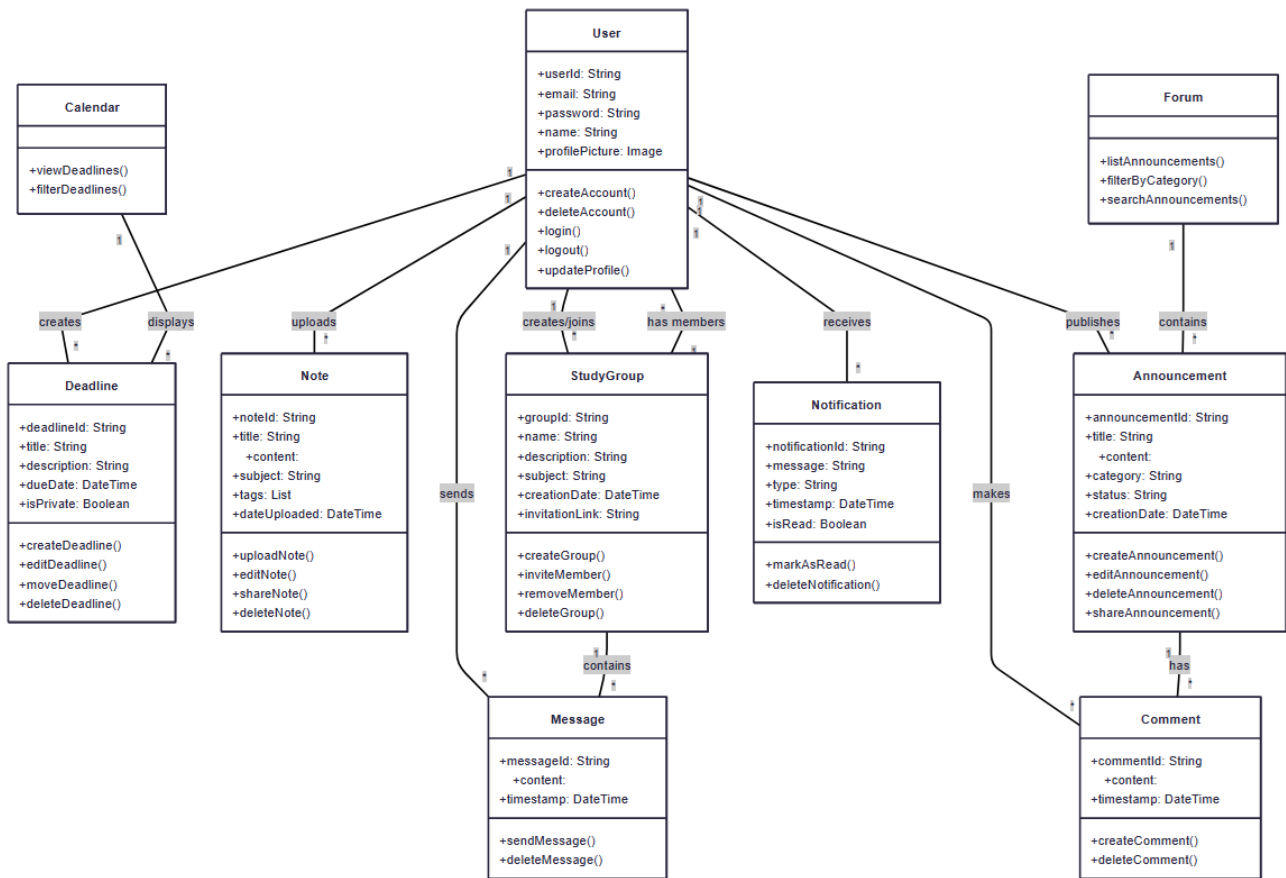


Figure 1: High level class diagram of the system

Domain Model Description:

This domain model represents the core structure of StudentHub, a platform designed to streamline student life management. The User entity is at the core of the system, with connections to all the major components. The Calendar interface allows users to create and manage deadlines, offering options for both private and shared academic tasks. The Note entity facilitates the uploading and sharing of study materials, organised by subject and tags.

StudyGroups facilitate collaboration between users, enabling members to exchange messages and coordinate study sessions. The creation of a group involves the sending of invitations to potential members. The Forum contains an Announcements section where users can post requests for books, offer tutoring services, or share event information. Should users wish to interact with announcements, they are able to do so through the 'Comments' function.

The model also includes a notification system that keeps users informed about deadlines, group invitations and forum activity. The relationships between entities reflect the platform's collaborative nature, demonstrating how users interact with content and each other.

This structure supports all of the identified goals of the RASD, including account management, deadline tracking, resource sharing, group collaboration, forum interactions, and marketplace functionality, creating a comprehensive ecosystem for student academic life management.

3 Specific Requirements

4 Effort spent

4.1 Argentini Elir

Task	Effort spent (hours)
Introduction	3
Overall Description	1:30
Specific Requirements	2
Reasoning	3
Writing and Fixing	3
Graphic Adjustments	0
Total	12:30

4.2 Battaglia Salvatore

Task	Effort spent (hours)
Introduction	2
Overall Description	1:30
Specific Requirements	2
Reasoning	3
Writing and Fixing	1
Graphic Adjustments	2
Total	11:30

4.3 Pantarotto Alessio

Task	Effort spent (hours)
Introduction	0
Overall Description	0
Specific Requirements	2
Reasoning	3
Writing and Fixing	3
Graphic Adjustments	1
Total	9

4.4 Rosiglioni Diego

Task	Effort spent (hours)
Introduction	0
Overall Description	0
Specific Requirements	1
Reasoning	2
Writing and Fixing	1
Graphic Adjustments	4
Total	8